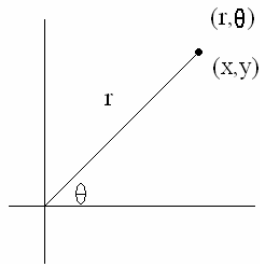


§10.3 Polar Coordinates

Homework : 5,13,17,25,39,55

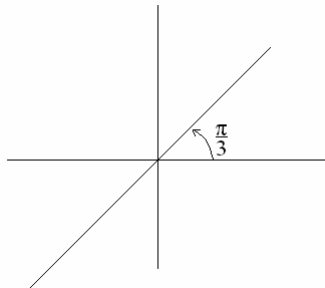
(I) 極座標



$(x, y) \leftrightarrow (r, \theta)$	
$x = r \cos \theta$	$r^2 = x^2 + y^2$
$y = r \sin \theta$	$\tan \theta = \frac{y}{x}$
i. r may be negative ii. 同一點 (r, θ) 的表法不唯一 $\left(-1, \frac{\pi}{4}\right) = \left(1, \frac{5\pi}{4}\right) = \left(1, \frac{13\pi}{4}\right)$	

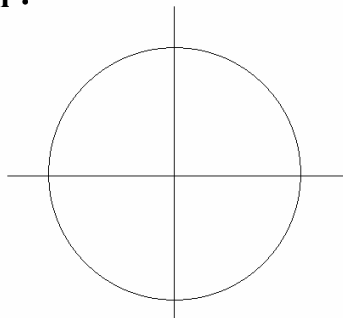
Example 1 : $\theta = \frac{\pi}{3}$ 代表直線.

Solution :

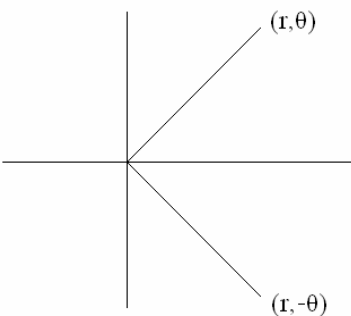
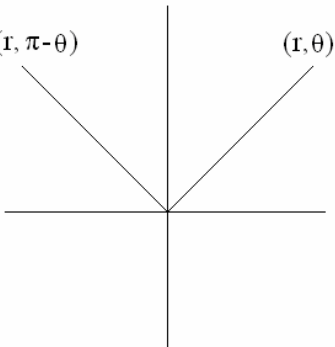
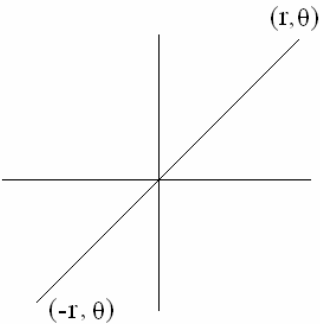


Example 2 : $r = 1$ 代表圓.

Solution :

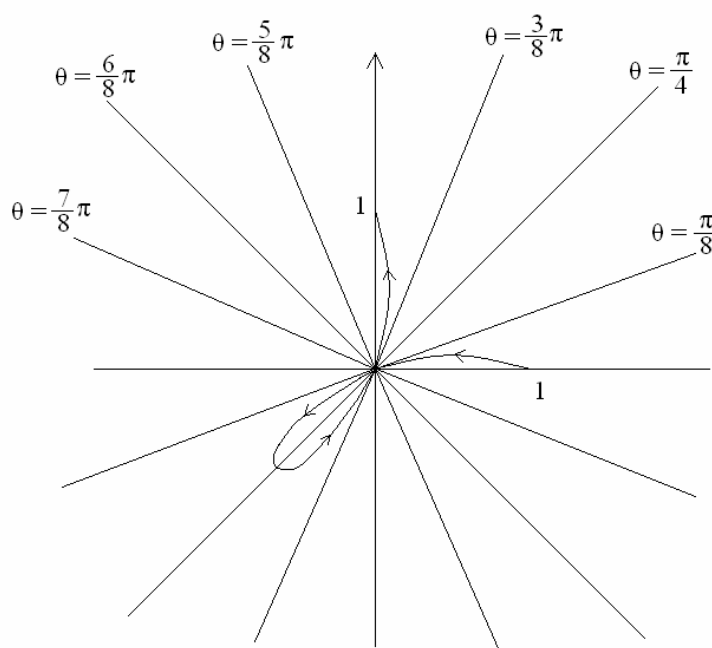


(II) Symmetry

對稱 x 軸	對稱 y 軸	對稱原點
θ 用 $-\theta$ 代 方程式不變	θ 用 $\pi - \theta$ 代 方程式不變	r 用 $-r$ 代 方程式不變
		

Example 3 : Sketch $r = 2 \cos 4\theta$

Solution : 對稱 x 軸和 y 軸



上題是 θ 從 $0 \rightarrow \frac{\pi}{8} \rightarrow \frac{2\pi}{8} \rightarrow \frac{3\pi}{8} \rightarrow \frac{4\pi}{8}$ (也即在第一象限) 的 r 的變化圖 (依箭頭方向)。

將此箭頭圖 reflect on the x -axis and y -axis 我們得到一個 8-leaved rose 的圖。

(III) Tangents to Polar Curves

Polar Curve : $r = f(\theta)$

Its corresponding parametric equation : $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$.

$$\text{Slope of the tangent : } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}.$$

Remarks :

i. Horizontal tangents : points at which $\frac{dy}{d\theta} = 0$.

ii. Vertical tangents: points at which $\frac{dx}{d\theta} = 0$.

Example 4 :

i. Sketch $r = 1 + \sin\theta$.

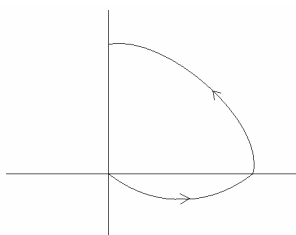
ii. Find $\left. \frac{dy}{dx} \right|_{\theta=\frac{\pi}{3}}$.

iii. Find the points on the curve $r = 1 + \sin\theta$ where the tangent line is horizontal or vertical.

Solution :

i. 對稱 y 軸，考慮 $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ 的部份

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$
r	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2



ii.

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = \frac{\cos \theta (1 + 2 \sin \theta)}{(1 + \sin \theta)(1 - 2 \sin \theta)}$$
$$\Rightarrow \left. \frac{dy}{dx} \right|_{\theta = \frac{\pi}{3}} = -1.$$

iii.

$$\frac{dy}{d\theta} = \cos \theta (1 + 2 \sin \theta) = 0 \quad \Rightarrow \quad \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}.$$
$$\frac{dx}{d\theta} = (1 + \sin \theta)(1 - 2 \sin \theta) \quad \Rightarrow \quad \theta = \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}.$$